CLINICAL · EAR

Keratosis obturans and canal cholesteatoma

A hard plug of keratin in the deep canal can be one of two conditions. In keratosis obturans the plug fills the canal all round, presses on it and widens it, painfully. In canal cholesteatoma the problem is a focal patch of eroded, dead bone on the canal floor. Both come from skin that has stopped migrating outwards.

AT A GLANCE

- Keratin plug: circumferential in keratosis obturans, focal in canal cholesteatoma.
- Clean regularly under the microscope.
- Biopsy granulation to exclude cancer.
- Surgery only for refractory disease.

1 CHAPTER ONE Overview

When the canal stops cleaning itself.

1.1 Normal self-cleaning

The skin of the drum and deep canal migrates outwards, carrying keratin and wax to the entrance of the canal. When this migration fails, keratin builds up in the deep canal.

1.2 Two conditions

KERATOSIS OBTURANS OR CANAL CHOLESTEATOMA?

	KERATOSIS OBTURANS	CANAL CHOLESTEATOMA
Keratin	Plug filling the whole canal	Focal, over an area of eroded bone
Bone	Widened all round, smooth	Focal erosion and necrosis, usually posteroinferior
Pain	Acute, severe	Dull, chronic, or none
Hearing	Conductive loss	Usually normal

	KERATOSIS OBTURANS	CANAL CHOLESTEATOMA
Sides	Often both	Usually one
Age	Younger	Older
Associations	Bronchiectasis, chronic sinusitis	Previous surgery, trauma, radiation

CHAPTER TWO

2 Keratosis obturans

The circumferential plug.

2.1 Cause

Failure of outward migration, possibly from inflammation of the canal skin. In young people it is associated with bronchiectasis and chronic sinusitis, which should be looked for.

2.2 Clinical picture

Acute, severe pain, conductive hearing loss and sometimes otorrhoea. A dense, white, laminated keratin plug fills the deep canal; once removed, the canal is widened and the drum may be thickened.

WHERE THE BOOKS DIFFER · AGE AND PAIN

Dhingra
Typically 5 to 20 years; pain is among the presenting symptoms.

Lee
Typically 30 to 60 years; severe pain is characteristic.

2.3 Treatment

Removal of the plug under the microscope, which may need a general anaesthetic in children, treatment of any secondary otitis externa, and regular cleaning. Salicylic

acid 2% in alcohol reduces reaccumulation. Follow-up is lifelong.

2.4 Complication

Progressive widening of the canal can erode into the mastoid (an *auto-mastoidectomy*) and expose the facial nerve.

3 CHAPTER THREE Canal cholesteatoma

Focal bone erosion.

3.1 Cause

Spontaneous in older patients, or after surgery, trauma, radiation or canal stenosis.

3.2 Clinical picture

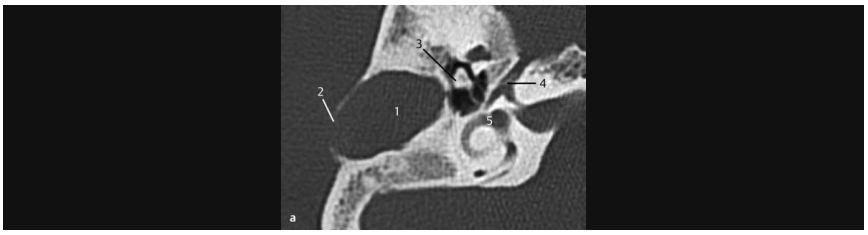
A dull, chronic ache and a foul, scanty discharge, with normal hearing. A localised ulcer on the floor or posterior wall of the bony canal, with keratin debris and exposed, necrotic bone.

3.3 Investigations

CT shows the extent of the erosion and its relation to the mastoid and facial nerve. Biopsy granulation tissue to exclude carcinoma.

IMAGING IN THE APP 1 case

Canal cholesteatoma with canal atresia.



Inclusion Cholesteatoma and Atresia of the External Auditory Canal
Congenital · 3 images

3.4 Treatment

Regular debridement and topical drops; when the disease keeps returning, surgery removes the matrix and dead bone and covers the defect with fascia or a skin graft.

OPERATIVE GUIDES IN THEATRE 2 operations

Keratosis Obturans and Canal Cholesteatoma
Keratosis obturans and canal cholesteatoma

Osteomas and Exostoses of the External Auditory Canal
Exostoses and osteomas

4 CHAPTER FOUR Differential diagnosis

What else causes a canal plug or erosion.

- Impacted wax.
- Necrotising otitis externa.
- Squamous cell carcinoma of the canal.
- Middle ear cholesteatoma eroding the canal wall.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Failure of epithelial migration** as the cause.
- **The comparison table:** circumferential versus focal.
- **Association with bronchiectasis and sinusitis.**
- **Biopsy** to exclude carcinoma.

5.2 Common pitfalls

COMMON PITFALLS

- Syringing a keratin plug repeatedly instead of removing it under the microscope.
- Missing carcinoma in canal granulation.

Written for this app from Oxford Textbook of Otolaryngology ch. 13, Oxford Specialist Handbook ch. 15, Oxford Handbook of ENT ch. 5, Dhingra ch. 8 and K. J. Lee ch. 22, 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

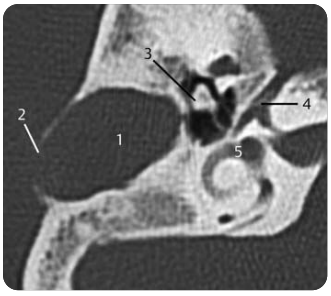
On imaging

The imaging of this topic: key findings, differentials and representative scans.

EXTERNAL EAR & TYMPANIC MEMBRANE · CONGENITAL

Inclusion Cholesteatoma and Atresia of the External Auditory Canal

- Expansion toward and/or destruction of the temporomandibular joint and toward the middle ear structures, abscess formation, and osteomyelitis and (intracranial) spread of infection.
- In case of aural atresia (first branchial groove anomaly), other dysmorphic features may be present too, especially in case of syndromal comorbidity.
- Particular attention needs to be paid to: the appearance of the middle ear cavity and mastoid pneumatization; signs of ankylosis or malformation of the ossicular chain; presence of inner ear deformities, the round and oval windows and the vestibular aqueduct; aberrations in the anterior and/or lateral course of the facial nerve, which may complicate surgery.



CT, axial. Expanding, round, smooth-bordered lesion (1) in the cranial part of the mastoid with partial destruction of the cortex (2). The head of the…